Analyzing Memory Effects in Large Language Models through the Lens of Cognitive Psychology

Zhaoyang Cao
Data Lab, Dept. of Electrical Engineering and Computer Science
Syracuse University
zycao@data.syr.edu

Lael Schooler
Department of Psychology
Syracuse University
lschooler@syr.edu

Reza Zafarani
Data Lab, Dept. of Electrical Engineering and Computer Science
Syracuse University
reza@data.syr.edu

arXiv:2509.17138v2 [q-bio.NC] 22 Oct 2025

Abstract

Memory, a fundamental component of human cognition, exhibits adaptive yet fallible characteristics as illustrated by Schacter's memory "sins".These cognitive phenomena have been studied extensively in psychology and neuroscience, but the extent to which artificial systems, specifically Large Language Models (LLMs), emulate these cognitive phenomena remains underexplored. This study uses human memory research as a lens for understanding LLMs and systematically investigates human memory effects in state-of-the-art LLMs using paradigms drawn from psychological research. We evaluate seven key memory phenomena, comparing human behavior to LLM performance. Both people and models remember less when overloaded with information (list length effect) and remember better with repeated exposure (list strength effect). They also show similar difficulties when retrieving overlapping information, where storing too many similar facts leads to confusion (fan effect). Like humans, LLMs are susceptible to falsely "remembering" words that were never shown but are related to others (false memories), and they can apply prior learning to new, related situations (cross-domain generalization). However, LLMs differ in two key ways: they are less influenced by the order in which information is presented (positional bias) and more robust when processing random or meaningless material (nonsense effect). These results reveal both alignments and divergences in how LLMs and humans reconstruct memory. The findings help clarify how memory-like behavior in LLMs echoes core features of human cognition, while also highlighting the architectural differences that lead to distinct patterns of error and success.

CCS Concepts

• **Computing methodologies → Cognitive modeling**.

Keywords

large language models, memory effects, cognitive psychology

1 Introduction

As the basis for learning, decision-making, and identity development, memory is an essential part of human cognition [3, 7, 29]. Yet, despite its vital role in daily life, memory is not always infallible when it comes to our experiences [24, 48]. Instead, it is a constructive process, vulnerable to illusion and error [5, 22, 39, 56]. Research in cognitive psychology has illuminated various complexities of memory, including its notable "malfunctions." Notably, Daniel Schacter [42–45] offers a taxonomy of seven memory "sins," which include *transience*, *absent-mindedness*, *blocking*, *misattribution*, *suggestibility*, *bias*, and *persistence*. Each of the sins reflects the adaptive qualities of human memory while also offering significant insights into its underlying constructive character. Various related fields in psychology, such as cognitive psychology [37], social psychology [12], and clinical psychology [50], along with cognitive neuroscience [8, 20, 41, 51], have conducted extensive studies on these memory errors. These errors are interpreted by adaptive byproducts, highlighting their evolutionary significance in human cognition and providing a framework not only to understand how memory malfunctions but also how it functions in complex environments.

Although human memory has been extensively studied, the emergence of Large Language Models (LLMs) provides a novel domain in which to apply and test the cognitive toolkit developed in memory research. Recent LLMs like GPT [1] and DeepSeek [23] can be regarded as large-scale associative memories that (1) retrieve knowledge through contextual search and (2) reconstruct information, perhaps in ways that are functionally similar to how human memory operates. These models, powered by the transformer architecture [54], have achieved great success in various applications, including but not limited to language translation [18, 34], text summarization [25, 33, 58], and even composing music and poetry [10, 17, 57]. Notably, the DeepSeek architecture (v2.3) introduces enhanced attention mechanisms and optimized training protocols that demonstrate superior memory retention capabilities compared to earlier models like GPT-3.5/4 [27, 28]. These systems process information through layered parameterized transformations inspired by biological neural plasticity, yet the extent to which they mimic human cognitive processes remains underexplored [31, 36, 52].

Benchmarks reveal that advanced LLMs like DeepSeek achieve superior performance on associative reasoning tasks compared to other state-of-the-art language models [27, 28]. These tasks encompass multiple cognitive dimensions, including but not limited to (1) multi-subject multiple-choice evaluation, (2) language understanding and reasoning, and (3) reading comprehension, among others. The observed performance improvements suggest that architectural innovations in models may better approximate human-like associative reasoning mechanisms in cognitive systems. This emerging capability raises an interesting and fundamental question grounded in cognitive science: *Do LLM memory systems exhibit Schacter's paradoxical 'sins' that demonstrate proper human memory functioning?* Specifically, (1) Which of these adaptive features are replicated in LLMs? and (2) Where do LLMs diverge from human memory systems, potentially revealing their mechanistic limitations? Such

analysis could establish whether LLM 'memory' operates via principles similar to human cognition, and these differences may help us understand how associative memory works in LLMs and suggest ways to improve their performance

Recently, Binz *et al.* proposed *Centaur*, a computational model capable of predicting and simulating human behavior [11]. They developed Centaur by fine-tuning on a large-scale dataset "Psych-101," which contains domains such as multi-armed bandits, decision-making, memory, supervised learning, Markov decision processes, and others. Their framework addresses early skepticism toward unified cognitive models by proposing a "cognitive decathlon" evaluation paradigm (a rigorous evaluation framework wherein sixteen experiments are used to test cognitive models, and their combined performance is evaluated), wherein Centaur outperformed established models across all sixteen rigorous experimental competitions. Centaur's success in simulating domain-general cognition validates data-driven approaches to cognitive modeling. Nevertheless, while Centaur focuses primarily on broad behavioral alignment, our research specifically investigates how large language models replicate or diverge from fundamental human memory phenomena.

Unlike the aforementioned research that employs LLMs to model the human cognitive processes, our study uses human memory research as a lens for understanding LLM's. Human memory has been the subject of meticulous research for 150 years (beginning with Ebbinghaus) that has produced numerous computational models of memory [19, 26, 32, 49, 53]. These models provide a framework and guide to understand the functioning of LLMs.

Our investigation evaluates experimental paradigms from human memory research - primarily associative recognition studies of the fan effect [2, 6] and the Deese-Roediger-McDermott (DRM) false memory [38] - to evaluate LLM memory architectures. We evaluate seven critical phenomena: list length effect (scaling performance with increasing memory load), list strength effect (interference in repetitive retrieval), fan effect (associative interference), nonsense effect (handling of non-meaningful stimuli), position effect (primacy/recency patterns in sequential memory), DRM-style false memories (false memory in a list learning paradigm), and cross-domain generalization ability (transfer of schema-consistent details). Our major contributions are summarized as follows.

- We conduct a systematic investigation of human-like memory effects (including list length/strength effects, fan effects, nonsense effect, position effect, and DRM-style false memories) across advanced LLM architectures, providing the information for evaluating progress toward human-level memory capabilities in LLMs.
- We demonstrate that LLMs exhibit similar characteristics of human memory effects: **i):** Human-like patterns in associative interference - fan effect. **ii):** Similar sensitivity to list length and list strength variations. **iii):** Parallel susceptibility to DRM-style false memories. **iv):** Comparable generalization ability. While LLMs mirror the qualitative patterns of human memory, they are more accurate, less affected by interference, and show stronger generalization, even as their pattern of errors remains similar to those of human cognition.
- Our experiments reveal where LLM memory systems fundamentally differ: **i):** Superior robustness to semantically anomalous content (nonsense effect). **ii):** Less influenced by the order in which information is encoded. All codes are available [1].

2 Theory Justification and Explanation

In research on memory and human cognition, various effects have been identified to describe how humans process and recall information. More specifically, memory retrieval can be broadly categorized into two aspects: recall and recognition. Recall refers to the active retrieval of information from memory in response to cues, reflecting a generative process. In contrast, recognition entails the identification of information as previously encountered when presented with a cue, relying on familiarity judgment. In this section, we first introduce the memory effects involved in the experiments in this paper. We will then present research that looks for patterns in large language models that parallel human recognition.

The ***list length effect*** captures the inverse relationship between memory load and recall performance. This phenomenon demonstrates that as the number of to-be-remembered items in a list increases, recall accuracy shows a decline, reflecting capacity limitations in long-term memory systems. [55]. The ***list strength effect***, first systematically investigated by Ratcliff, Clark, and Shiffrin, represents a fundamental characteristic of competitive memory retrieval. This phenomenon manifests in two distinct ways: in free recall, strengthening certain items in a memorized list (typically through repetition or extended study time) harms the remaining list items; in recognition, it appears as a missing or negative effect [35, 47]. The ***fan effect*** proposed by Anderson describes how an increase in associations to an item leads to slower retrieval times for information associated with the item [6]. Formally, a fan refers to the number of associations of a concept in a semantic network. Schneider and Anderson extended this finding by demonstrating two additional observations: (1) high-fan items exhibit reduced asymptotic accuracy (the theoretical maximum recognition accuracy as study times approach infinity), and (2) their accuracy growth rate - quantified as the derivative of accuracy over lag - is significantly slower compared to low-fan items [46]. The ***nonsense effect*** describes the robust disadvantage in memorizing semantically incoherent material (e.g., random words or non-words) compared to meaningful information (e.g., paragraphs or associative word pairs). This effect arises from a key mechanism: the lack of semantic scaffolding, where existing knowledge structures cannot support encoding or retrieval [16]. The ***Deese-Roediger-McDermott paradigm*** (known as the DRM effect) illustrates how people recall related but non-presented words in a list, revealing the impact of semantic activation on false memory. When participants study lists of strongly related words (e.g., thread, pin, eye, sewing for the critical lure 'needle'; or bed, rest, tired, dream for 'sleep'), they often falsely recall or recognize the non-presented lure word [14, 15, 30]. This occurs because semantically related words activate the lure concept in memory, which causes it to be incorrectly recalled alongside elements that are actually presented [38]. Additionally, the DRM effect is often used as a model laboratory task for the formation of false memories in people. The ***position effect*** captures a fundamental property of sequential information processing: items at

[1]Data and analysis code available at https://github.com/zycao29/LLM_CognitivePsychology.git.

the beginning (*primacy effect*) and end (*recency effect*) of an input sequence are generally remembered better than middle items [4, 5]. This effect occurs in both artificial and biological memory systems and was initially measured in controlled memory studies when participants recalled word lists in presentation order [47]. Lastly, ***generalization ability*** in memory systems reflects a fundamental tradeoff: while precise memory maintains accurate details, weaker memory traces often lead to broader and consistent generalizations. This phenomenon is empirically demonstrated through the following landmark cognitive experiments: **Text Memory Paradigm**: When recalling a story about "a doctor treating patients," individuals with weaker verbatim memory tend to reconstruct the core meaning while substituting schema-consistent details (e.g., changing "neurosurgeon" to "doctor"), whereas those with stronger memory preserve specific terminology but show less conceptual flexibility [9, 13]. Collectively, these effects serve as benchmarks to assess the alignment of memory dynamics in human cognition with large language models.

3 Dataset Modification

Our experimental framework leverages two datasets to systematically evaluate memory phenomena in large language models. The first dataset ("Person Location Lists", see Appendix Table S1) builds upon person-location pairs from Schneider and Anderson's paper [46], which we augmented with additional words to enable comprehensive testing of the ***list length/strength effects, fan effect***, and ***nonsense word effect***. The second dataset "Target words and Associates Lists" (see Appendix Table S2) comprises carefully selected word association lists derived from [38], specifically chosen to assess ***DRM-style false memory formation, position effect***, and ***generalization ability***.

Dataset: "Person Location Lists"

The dataset is organized into four controlled sub-experiments (Experiments 1-4) that evaluate fundamental memory effects: Experiment 1 examines memory capacity via list length variations, Experiment 2 manipulates item strength to test competitive retrieval, Experiment 3 targets associative interference through the fan effect paradigm, and Experiment 4 evaluates nonsense effect. Specifically, we prepared (I) a study list, (II) a foil list, and (III) a question list. The study list contains a series of facts, each following the form "`The <person> is in the <location>.`" These were generated from sets of 40 persons and 40 locations. The word lengths for persons and locations ranged from 3 to 12 characters (mean = 6.85, sd = 1.93) and from 4 to 10 (mean = 6.33, sd = 1.83), respectively. The size of the study list varied according to each sub-experiment. The foil list included non-studied facts created by rearranging persons and locations while maintaining the *fan status* (word frequency) of the study list. The question list covered both the study and foil facts, converting their facts into question format using the form "`Is the <person> in the <location>?`". The first half of the question list contained questions from studied facts, while the second half contained non-studied facts.

Dataset: "Target Words and Associates Lists"

The dataset is organized into three additional controlled sub-experiments (Experiments 5-7) that evaluate distinct memory phenomena: Experiment 5 examines recall patterns via serial position effects, Experiment 6 investigates false memory formation through the Deese-Roediger-McDermott paradigm, and Experiment 7 assesses memory flexibility through cross-domain generalization ability. Specifically, we selected 12 target words for the experiment, and for each target word, 15 associated words were selected for the list. The words in the list were ordered based on the strength of their association, with the most strongly associated words appearing first. The selection and ordering of target words and their associates were referenced by the lists provided in Roediger's study [38]. The finalized lists are available in the Appendix Table S2.

4 Results

We conducted experiments with various large language models, including GPT-4, Mistral, and LLaMA2, and observed consistent trends across these models. While some models demonstrated superior performance in specific memory-related tasks, none of the evaluated LLMs achieved perfect results across all memory experiments. Notably, GPT-4 exhibited the most robust performance among the tested models. Therefore, we primarily present the detailed results obtained from GPT as it consistently outperformed other models in our comparative analysis.

Dataset 1 "Person Location Lists"

The results on Dataset 1 are shown in Figure 1. Figure 1 leads us to the following conclusions. 1: Regarding the fan effect, we can observe that when the fan value increases, the recall accuracy of the language model for the recognition test list shows a negative relationship; for example, the value decreases from 0.991 to 0.915 when the list size is 32. The trend of the results showed that increasing the number of associations to an item results in slower retrieval. 2: With respect to the list length effect, we can observe that the recall accuracy of the language model decreases as the list size (the number of studied facts) increases, even though there are some fluctuations. 3: In terms of the list strength effect, although the relationship is not particularly significant in the first group, our results demonstrate that stronger items should yield higher accuracy than weaker items, while the accuracy of weaker items remains unaffected by the presence of stronger ones. Consequently, the observation is that groups excluding repeated items exhibit slightly lower accuracy. 4: Regarding the nonsense effect, whether replacing one type of word alone or replacing both person and location does not seem to have a significant impact on the recognition accuracy of the language model.

Dataset 2 "Target Words and Associates Lists"

Table 1 illustrates the results on Dataset 2. We can find that, whether it is an immediate recall or a later recall (by doing math problems), the hit rate (i.e., the recall accuracy for those studied words) is always 1. In addition, for non-studied words, the false alarm rate of the language model is 0.053. This indicates that language models may incorrectly recognize words that were not present in the study list. The results are highly consistent with the DRM effect observed